

Lakshmi Sahasra Jangoan

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github.com/jangoansahasra | Portfolio

PROFESSIONAL SUMMARY

Data professional with 2.5+ years of experience spanning data analytics, data science, and data engineering, delivering high-impact ML models, BI dashboards, and production-grade data pipelines from large, real-world datasets. Proficient in SQL, Python, and R for statistical modeling, predictive analytics, and time series forecasting (XGBoost, LSTM, ARIMA), and in Power BI, Tableau, and DAX for building dashboards and KPIs that drive stakeholder decisions. Experienced in ETL/ELT pipeline design, data validation, and API/cloud deployment (FastAPI, Docker, Azure, PostgreSQL) for scalable, quality-assured data systems, with hands-on exposure to deep learning and GenAI/LLM applications. Proven ability to frame ambiguous problems, catch data quality issues early, and communicate findings to technical and non-technical stakeholders alike.

TECHNICAL SKILLS

Languages & Querying: Python (Pandas, NumPy, Scikit-learn, PyTorch), SQL (CTEs, Window Functions, Joins, Subqueries, Aggregations), R, MATLAB, JavaScript, Bash

BI, Analytics & Visualization: Power BI (DAX, Data Modeling), Tableau, Excel (Pivot Tables, VLOOKUP/XLOOKUP), Dashboard Development, KPI Reporting, Ad-hoc & Exploratory Data Analysis

Machine Learning & AI: Regression, Classification, XGBoost, Random Forest, Deep Learning (TensorFlow, PyTorch, ResNet, LSTM/GRU), Time Series Forecasting (ARIMA), NLP, LLMs & RAG, Hypothesis Testing, Cross-Validation

Data Engineering & Cloud: ETL/ELT Pipeline Design, PostgreSQL, Azure, FastAPI, Docker, RESTful APIs, Data Validation & Quality Assurance, Git, GitHub, Pytest, HPC/Distributed Computing

PROFESSIONAL EXPERIENCE

Data Scientist - Research | *National Weather Center, University of Oklahoma* | *Norman, OK* Oct 2024 – Jun 2026

- Built end-to-end ML pipelines for precipitation forecasting using 40+ years of ERA5 reanalysis data across 156 atmospheric channels, combining ResNet18-based deep learning feature extraction with ensemble and autoregressive modeling to improve forecast accuracy by 14% over baseline.
- Engineered temporal lag/lead and spatiotemporal features across 3D climate data formats; executed 600+ model training runs on HPC systems using PyTorch, Scikit-learn, and xarray, reducing RMSE by 18% and MAE by 13% versus conventional approaches.
- Performed model diagnostics including calibration analysis, anomaly detection, and distribution-aware evaluation, translating complex results into dashboards and reports for 5+ cross-functional research stakeholders.

Teaching Assistant - Time Series Analysis | *University of Oklahoma* | *Norman, OK* Dec 2025 – May 2026

- Mentored 40+ graduate students on ARIMA modeling, regression, and statistical hypothesis testing using Python, MATLAB, and R, translating complex time-series methods into applied analytical techniques.
- Graded and delivered detailed feedback on 100+ assignments and projects, identifying recurring analytical errors and raising cohort proficiency in model evaluation.

Data Analyst | *ValueLabs* | *Hyderabad, India* Nov 2023 – Jul 2024

- Analyzed 120,000+ operational records using complex SQL queries (CTEs, window functions, joins, aggregations) to clean, validate, and standardize data, improving accuracy by 15%.
- Designed and deployed 10+ Power BI dashboards with KPI tracking and drill-through filters, cutting manual reporting effort by 25% and accelerating stakeholder decision-making.
- Partnered with cross-functional stakeholders to translate ambiguous business questions into structured data models and analytical reports supporting operational decisions.

PROJECTS

Public Health Outbreak Forecasting | *Python, Time Series, XGBoost, LSTM, ARIMA, Tableau*

- Contributed to an OU-UNSA international public health research grant, building ARIMA, XGBoost, and LSTM models on 20+ years of surveillance data to forecast respiratory infection outbreaks in Peru, achieving 0.94 R^2 and a 12% RMSE reduction.
- Built Tableau dashboards translating outbreak trends for non-technical stakeholders, cutting reporting time by 35% and supporting public health monitoring decisions.

Enterprise Urban Mobility Data Platform | *Python, PostgreSQL, FastAPI, Docker, XGBoost, ETL*

- Designed an end-to-end ETL pipeline ingesting and transforming 10.4M+ NYC taxi trip records into a PostgreSQL star schema, achieving 100% data-quality compliance across all validation checks.
- Engineered FastAPI REST endpoints serving dual XGBoost models (R^2 up to 0.91) for trip-duration prediction, implemented automated Pytest tests, and containerized the platform with Docker.

Local AI Assistant with RAG Pipeline | *Python, FastAPI, Ollama, ChromaDB, LLMs*

- Built a fully local retrieval-augmented generation (RAG) pipeline running Llama 3.2, Phi-4 Mini, and Mistral 7B via Ollama with ChromaDB and sentence-transformer embeddings, reaching 45 tokens/sec, 2.4x faster than baseline on 53% less memory.
- Designed FastAPI REST endpoints and a Streamlit interface with prompt engineering to improve response relevance, demonstrating deployable, cost-free GenAI tooling.

Energy Consumption Forecasting & Optimization (Steel Manufacturing) | *Python, NumPy, Regression, K-Means, Streamlit*

- Developed 3 core data mining algorithms entirely from scratch (regression, K-fold cross-validation, K-Means) on 35,040 industrial energy records, integrating them into a unified forecasting and monitoring pipeline.
- Built a statistical anomaly detection module (Z-score/IQR) that flagged 2,119 abnormal energy events (6.05% rate, peak 157 kWh), delivered through an interactive Streamlit dashboard for real-time operational monitoring.

EDUCATION

University of Oklahoma (GPA: 4.0/4.0)

Master of Science, Data Science and Analytics

Aug 2024 – May 2026

Norman, OK

Jawaharlal Nehru Technological University

Bachelor of Technology, Computer Science and Engineering

Dec 2020 – May 2024

Hyderabad, India